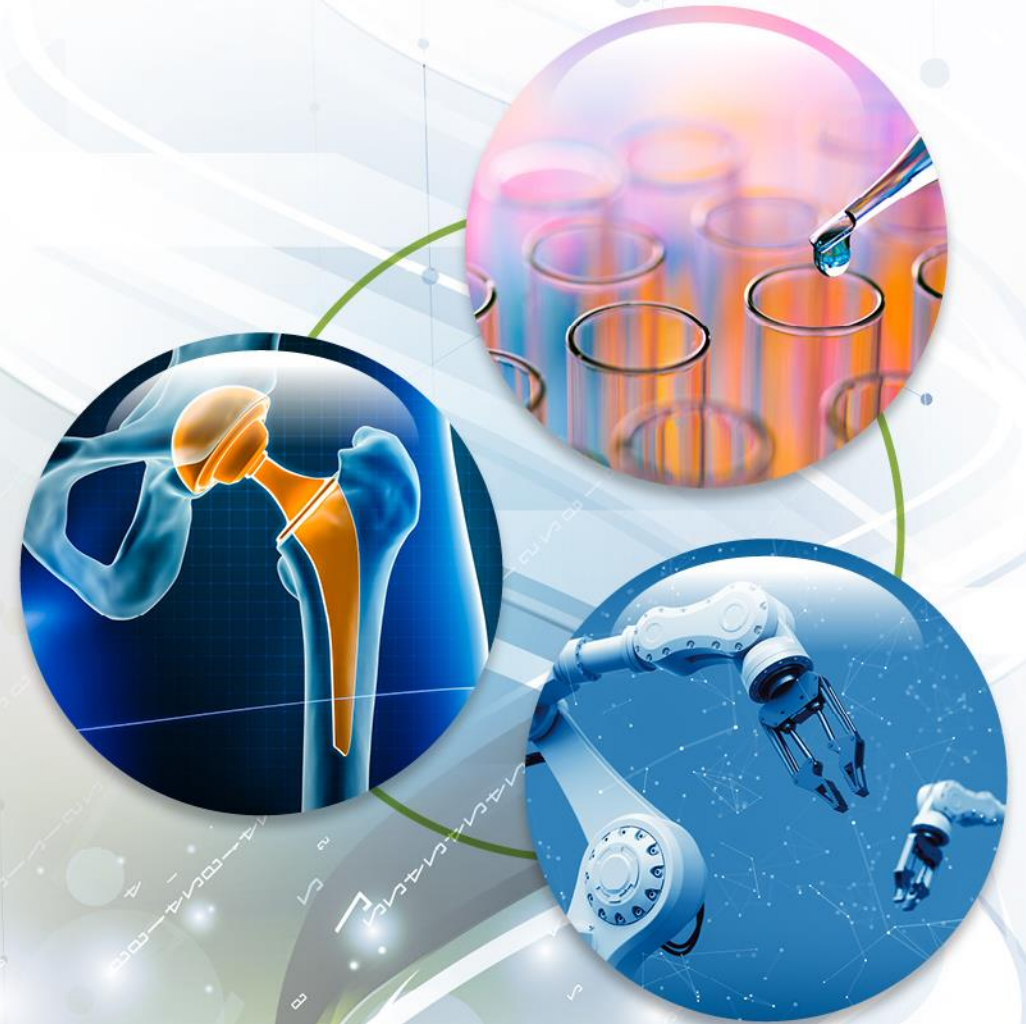


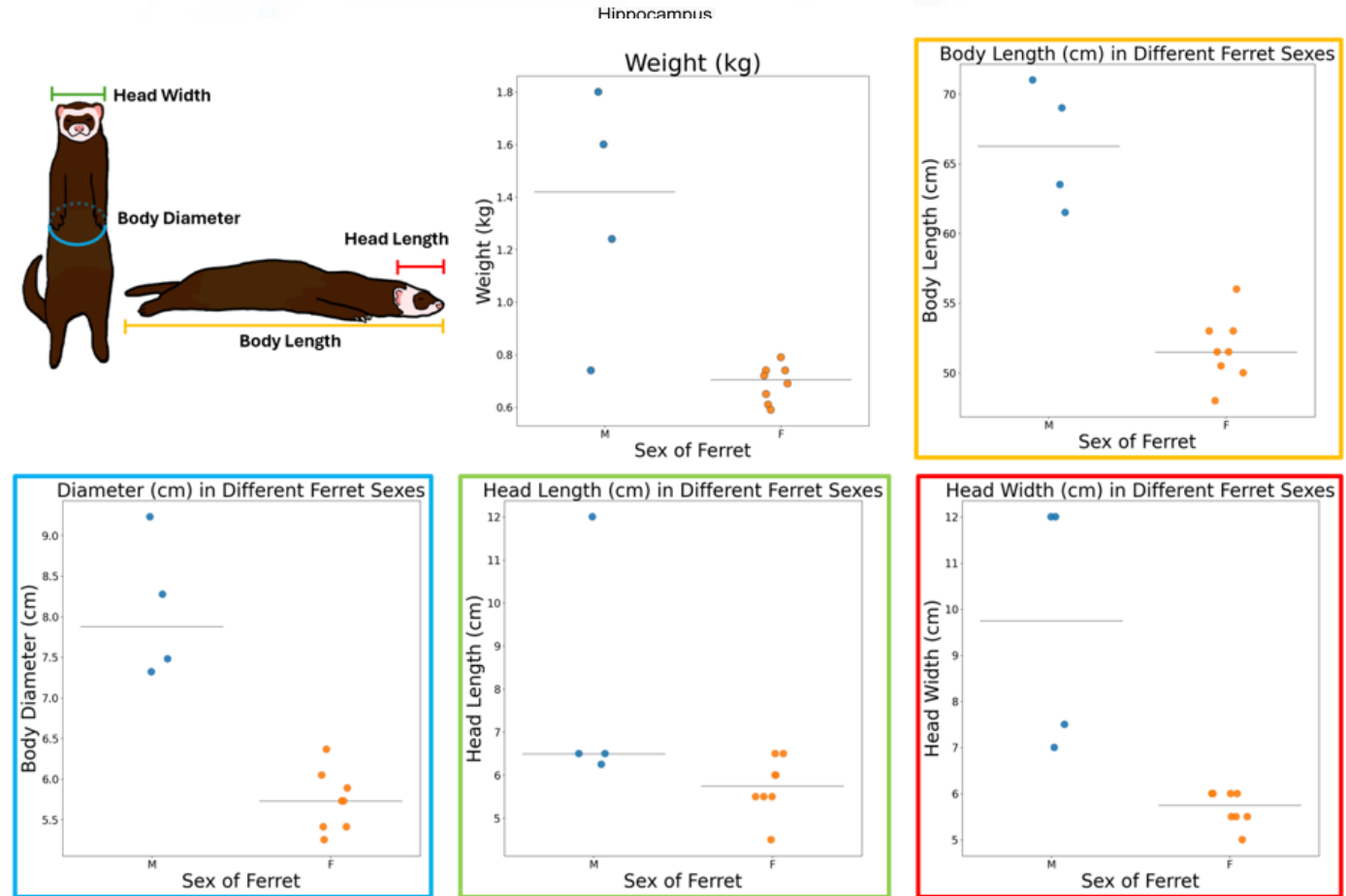
Adaptable 3D-Printed MRI Cradle for Ferret Brain Magnetic Resonance Imaging with Integrated Anesthesia and Commercial Coil Support

Alvaro Cruz Perez, Laurel Dieckhaus Jesus Mendoza, Leili Falbinan, Diego Celdran-Bonafonte, Katy Phillip, Rhea Carlson, Jean-Phillipe Galons ,Elizabeth Hutchinson



Background

- Ferrets are a promising pre-clinical model for brain MRI studies
- Size variance: specifically sexual dimorphic differences in ferrets
- There is no a commercial solution
- Improve previous setups for ferret brain MRI
- 3D-printed custom-made solutions



Previous setups

- Transmit-receive volume coils with inner diameter of 60mm and 86 mm (Hutchinson, E. B. et al., 2017)
- Rat cardiac coil and anesthesia delivery system (Celdran-Bonafonte et al., 2025)

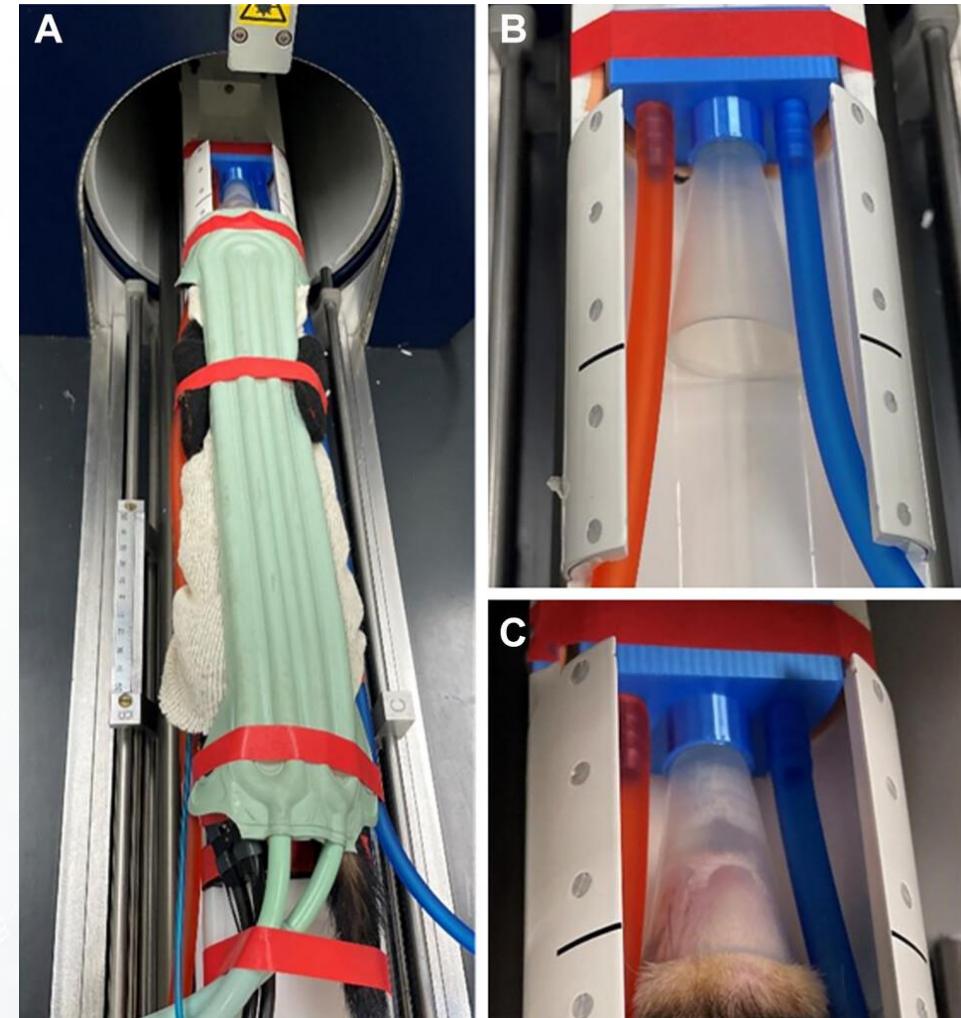
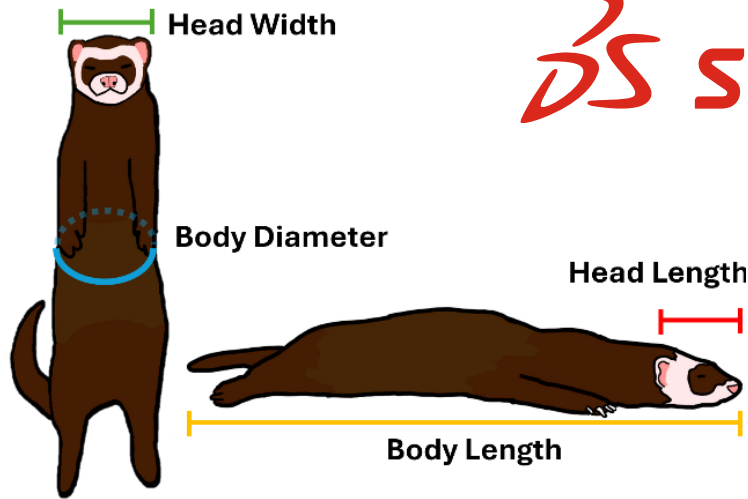


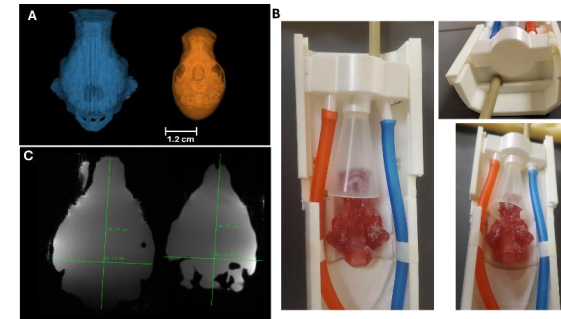
Figure 3. Ferret setting for brain MRI. (A) Note the direct heating element (circulating water system). (B and C) Note the custom 3D-printed inhaled anesthetic delivery system. (Celdran-Bonafonte et al., 2025)

Design Process and Testing

Feedback from
MRI and
veterinary staff



SOLIDWORKS



Ferret
imaging

BMES
BIOMEDICAL ENGINEERING SOCIETY



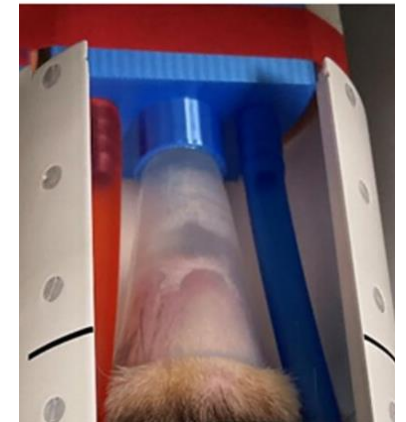
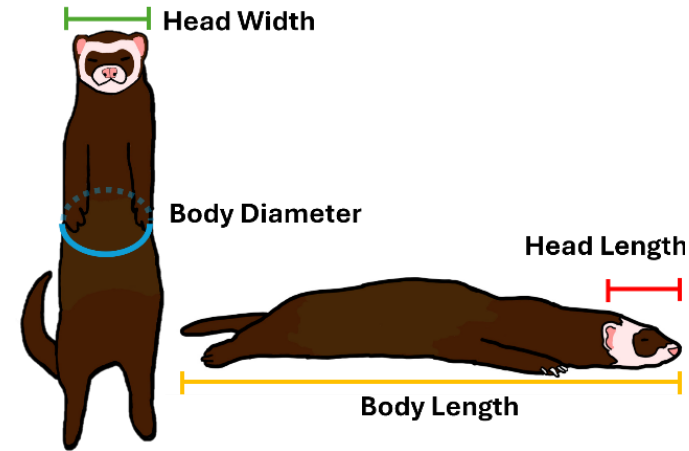
Design requirements

Size Requirements

- Must fit the largest male ferret (~ 50 cm. ~10 cm)
- Must adjust to different sizes (~ 30 cm, ~ 5 cm)

Veterinary Requirements

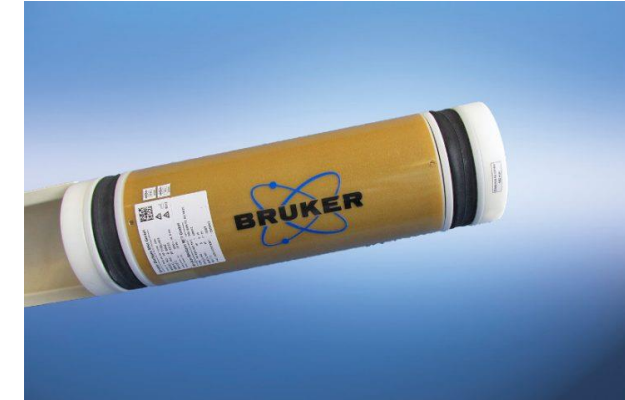
- Must integrate an anesthesia system
- Must have easy access to the body
- Must provide space for heating pad/blankets



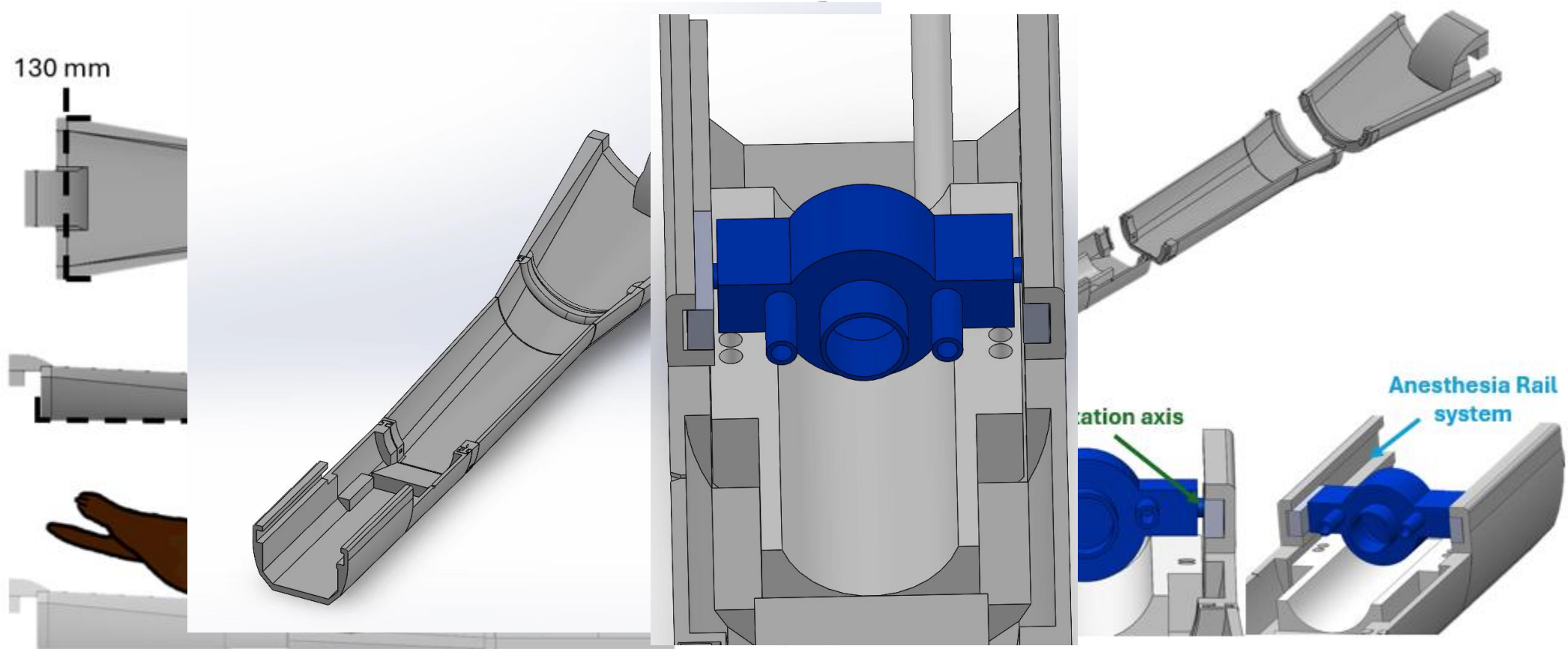
Design requirements

MRI System Requirements

- Must use the 86 mm cylindrical coil and rat brain surface coil in conjunction
- Must reach the center of the 86mm coil
- Must be made of MRI-compatible material
- Must deliver a similar field of view and SNR
- Must fit the mounting fixture on the scanner
- Should be modular

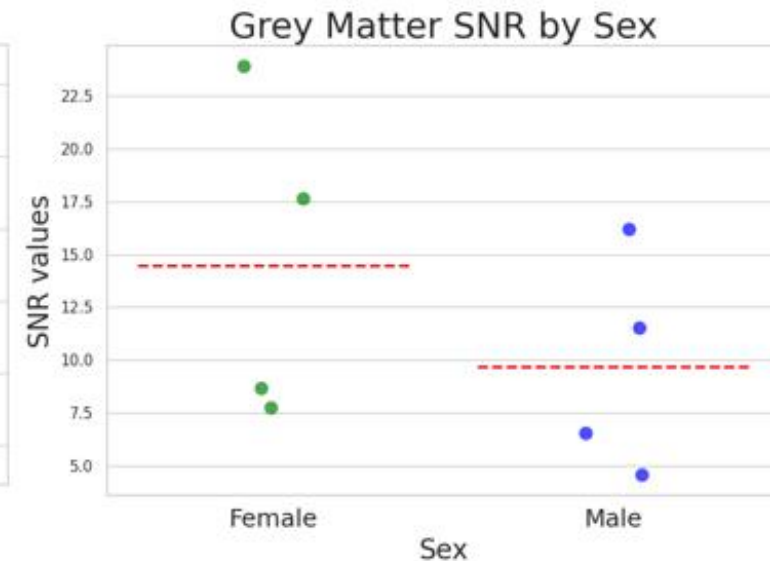
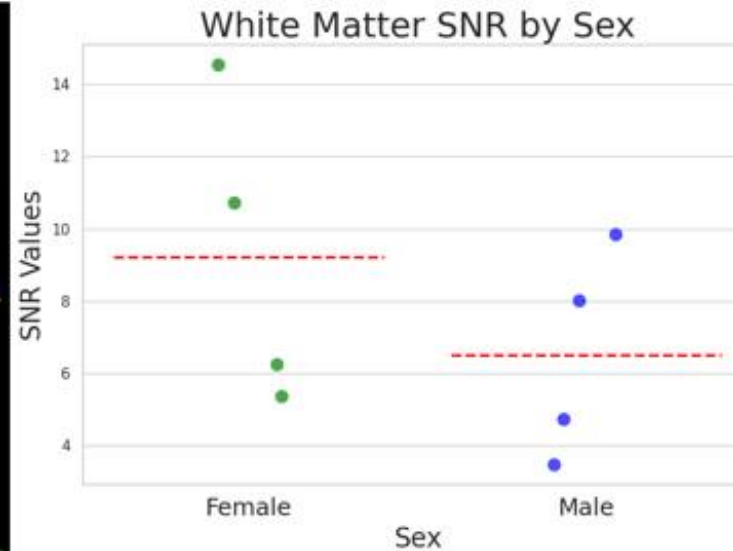
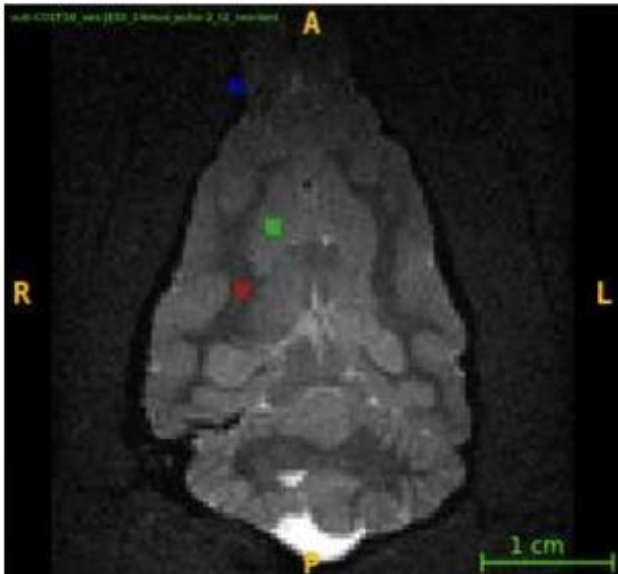
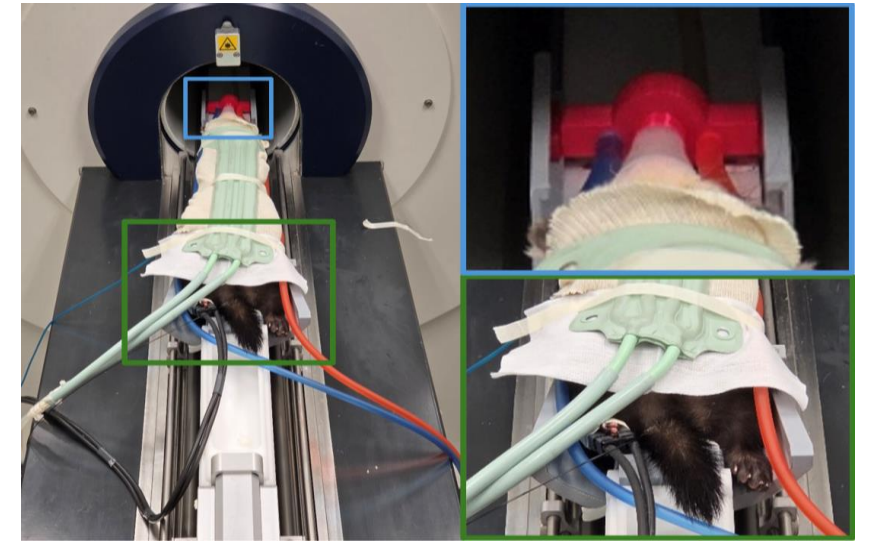


Design features – PLA +



Successful In-vivo imaging

- In-vivo imaging was performed in 8 ferrets of different sizes (4 males and 4 females)
- NO susceptibility or motion artifacts
- The SNR comparison from the in-vivo imaging between the new cradle and the previous female setup show that there is no significant difference ($p > 0.05$)



Conclusions

- Veterinary staff observations, multiple imaging sessions and SNR analysis proved that the cradle is suitable to image ferrets of different sizes
- 3D printing offers an easy, fast prototyping and low-cost option to make MRI-compatible tools to improve imaging methods

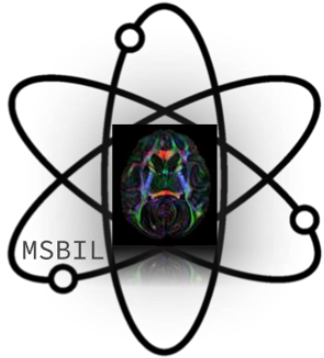
Future Directions

- Longitudinal studies for both males and females

Limitations

- Signal strength decays with distance
- Tolerances of 3D printing

Acknowledgements



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Veterinary collaborators:
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Katy Phillip

MSBIL Website



References



Laurel A. Dieckhaus
Postdoc

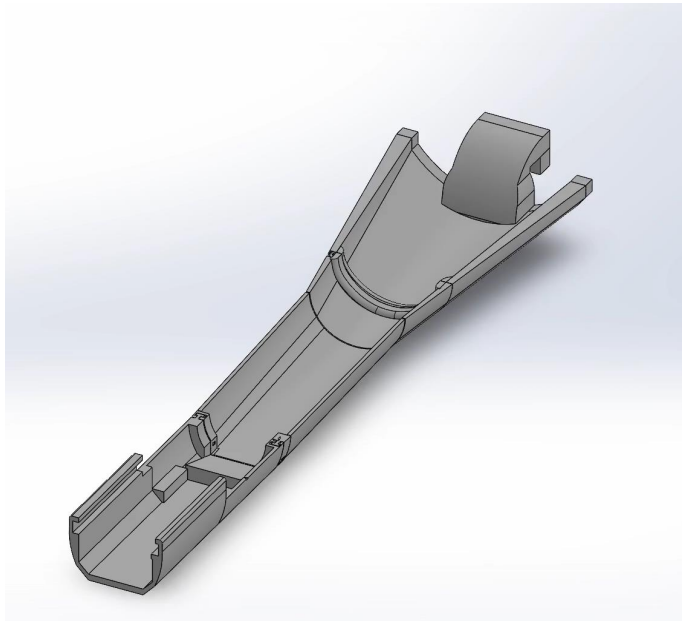


Elizabeth B. Hutchinson
Principal Investigator

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Any questions?



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Printing settings and other materials

- Nylon
- Generic PLA

Printer: Bambu lab H2D

- Infill: 30% (infill pattern grid)
- Supports: Standard (threshold angle 30)
- Layer height: 0.2mm
- Extruder nozzle temperature : 220C
- Printing plate: Textured PEI temperature:55C
- Max volumetric extrusion : 25mm³/s
- Min speed: 20mm/s

